

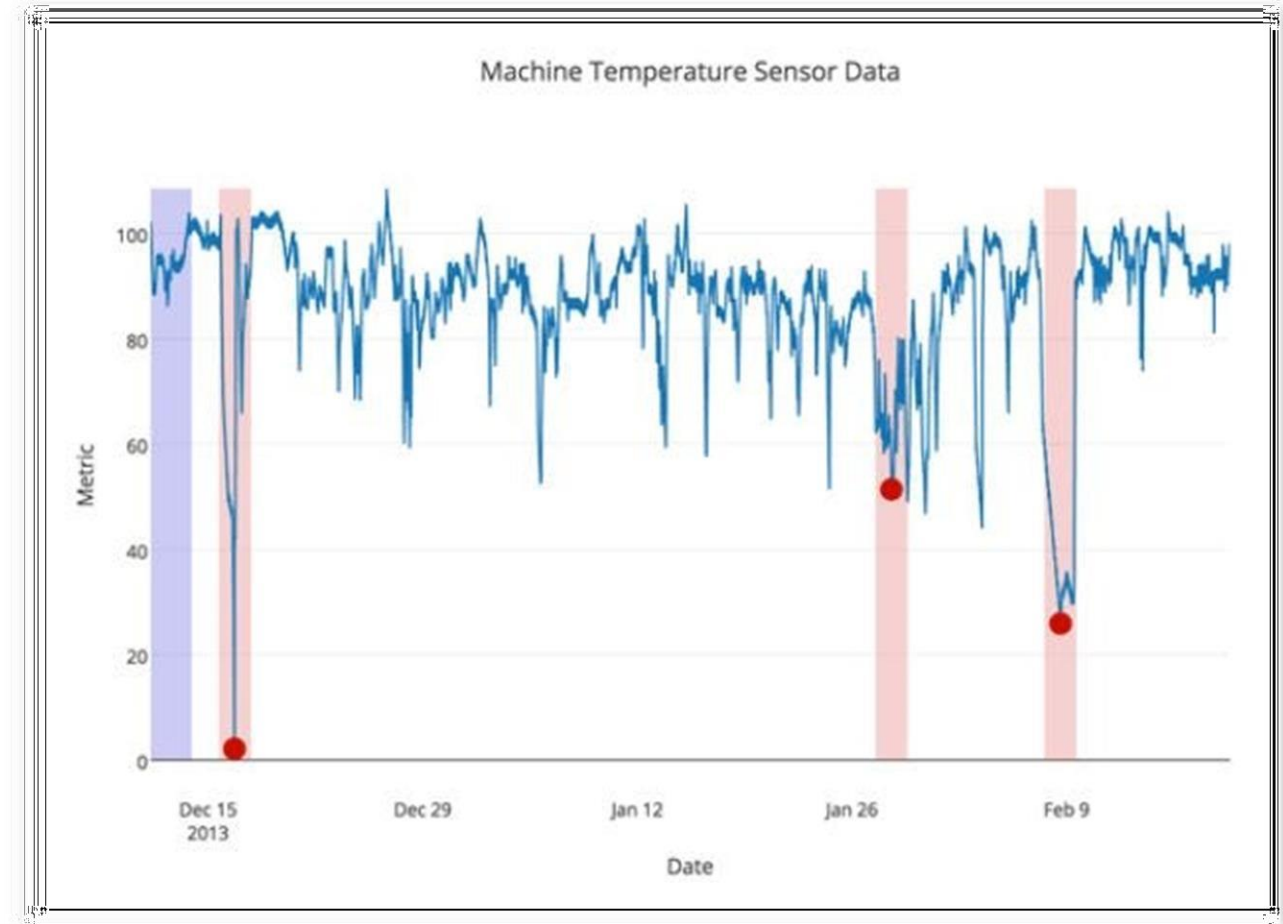
IoT Capstone Project Report

Temperature IoT Data Anomaly Detection (Using ML methods)

Semester: Fall 2025

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Instructor: Dr. Neha Yadav



Applications of Anomaly Detection in IoT

Enhancing IoT Security and Efficiency

Industrial Automation:

Early identification of equipment malfunctions or overheating in industrial settings.

Energy Management:

Optimizing energy consumption by identifying anomalies in temperature patterns in smart buildings and facilities.

Healthcare:

Ensuring proper storage conditions for pharmaceuticals and medical supplies.

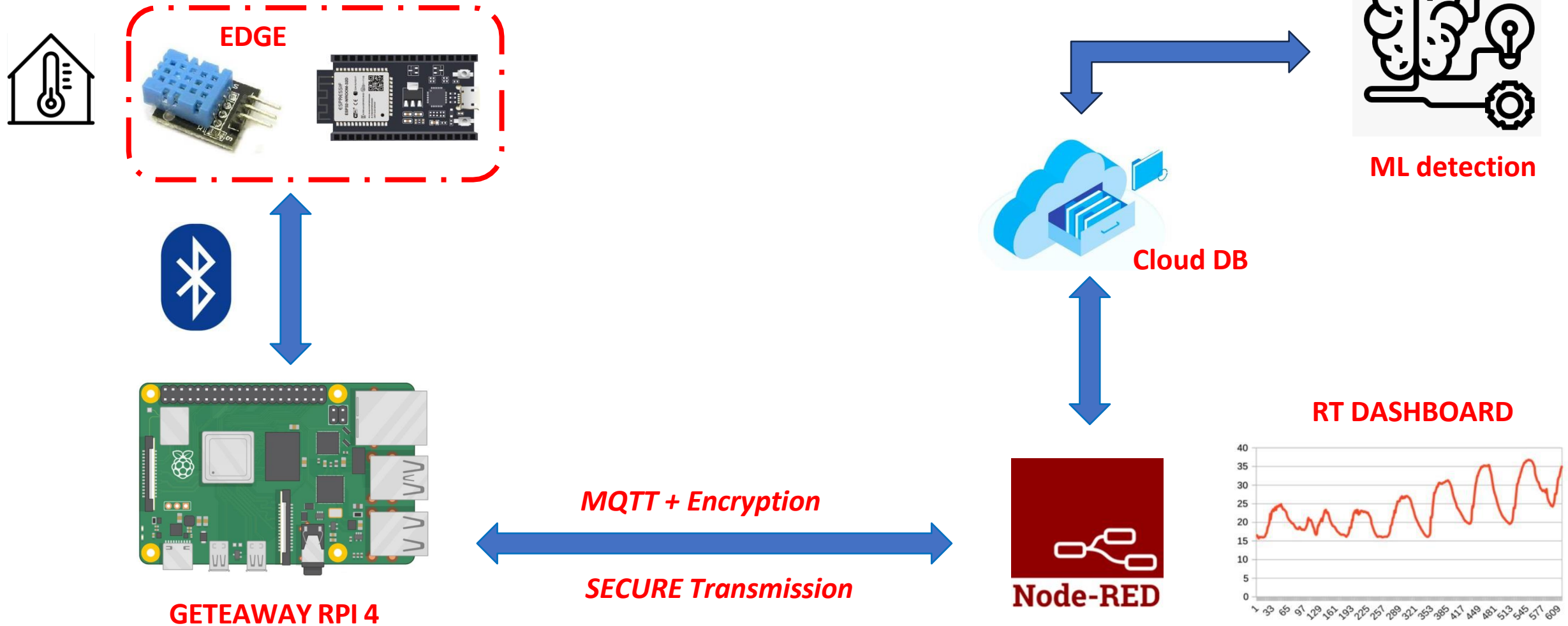
Agriculture & Food Safety:

Early detection of climate-related anomalies affecting crop growth.

Ensuring the correct storage temperature for perishable goods during transportation and storage.

System Diagram

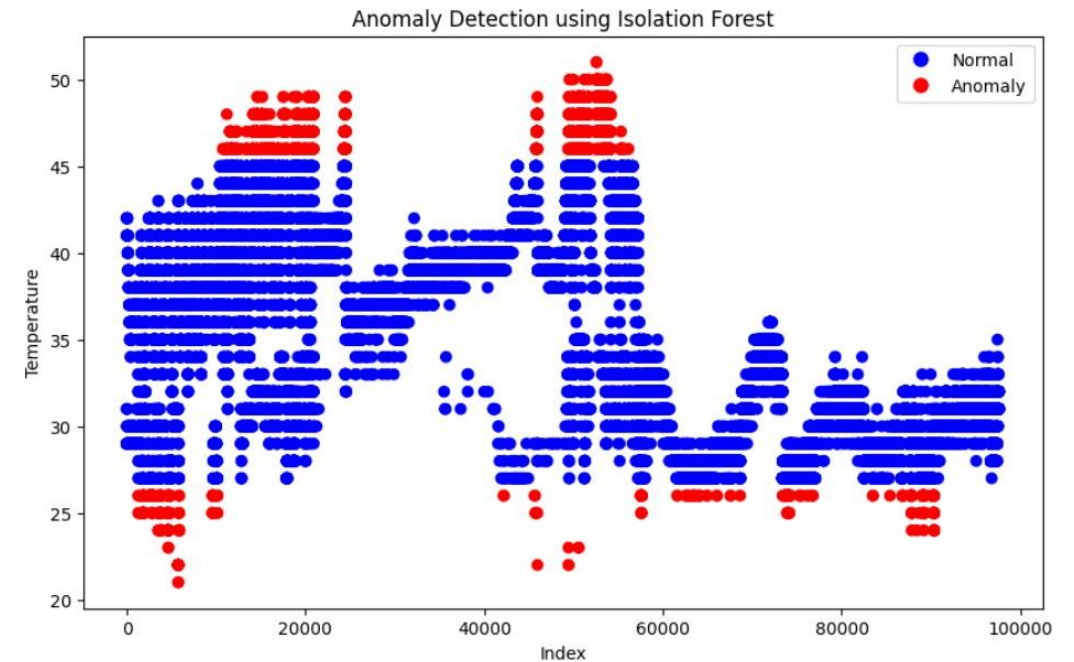
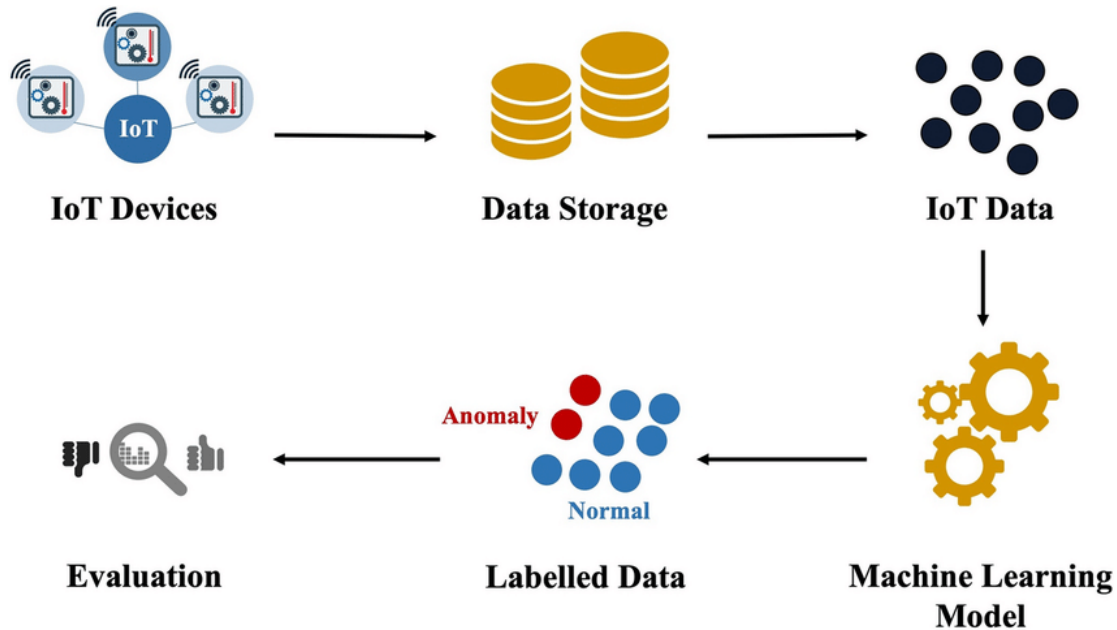
Overview of IoT System Components



Review:

Steps for Implementing Anomaly Detection in IoT

From Data Collection to Actionable Insights

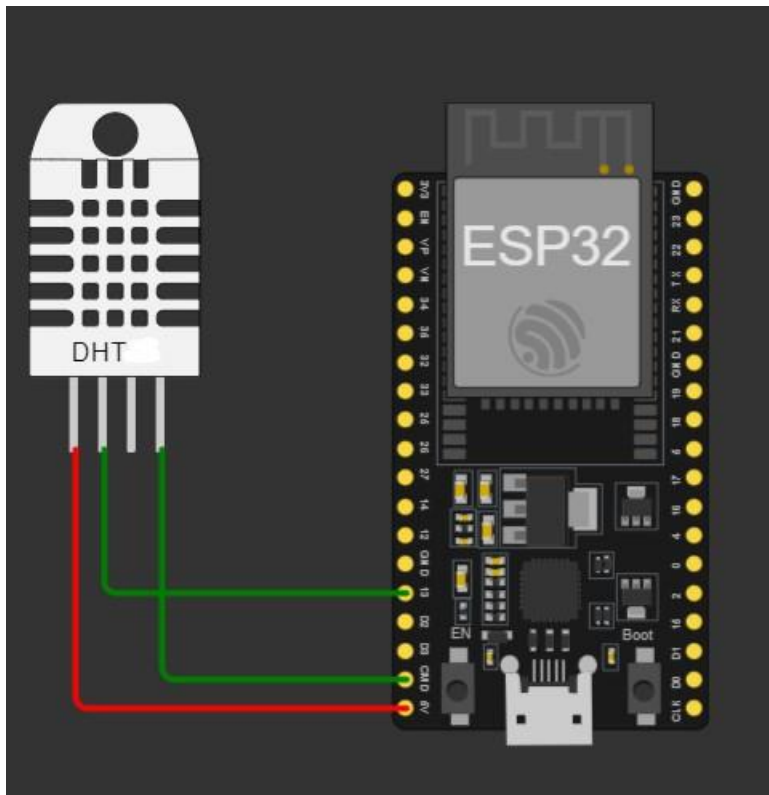


e.g. Isolation Forest

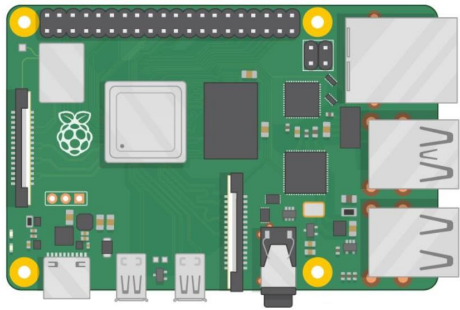
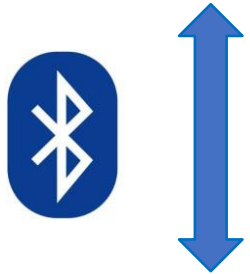
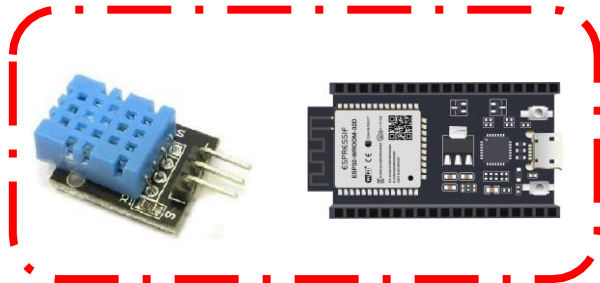
Part 1: (Design / Hardware / Sensors)

Connecting the DHT11 Sensor to ESP32 and Receiving the temperature:
After Installation of the sensor with the following format, the script to show the temp is used.

The Schematics:



Part 2: (Communications)



Sending the received data via Bluetooth to Raspberry pi. (C Code)

Receiving the data via Bluetooth in Raspberry pi. (Python Code)

The bluepy library is used

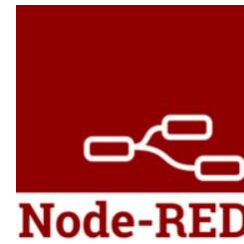
In the part the root access is required.

Receiving the data via Bluetooth in Raspberry pi,

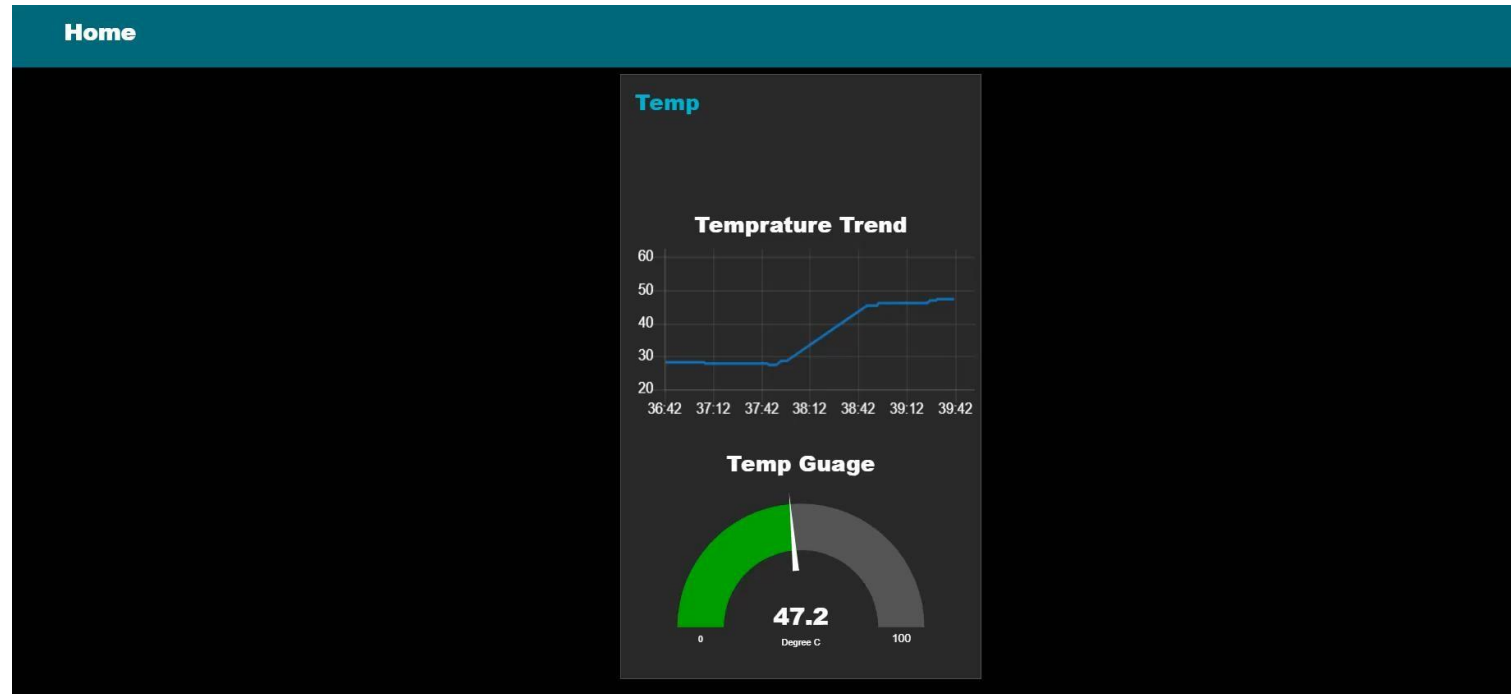
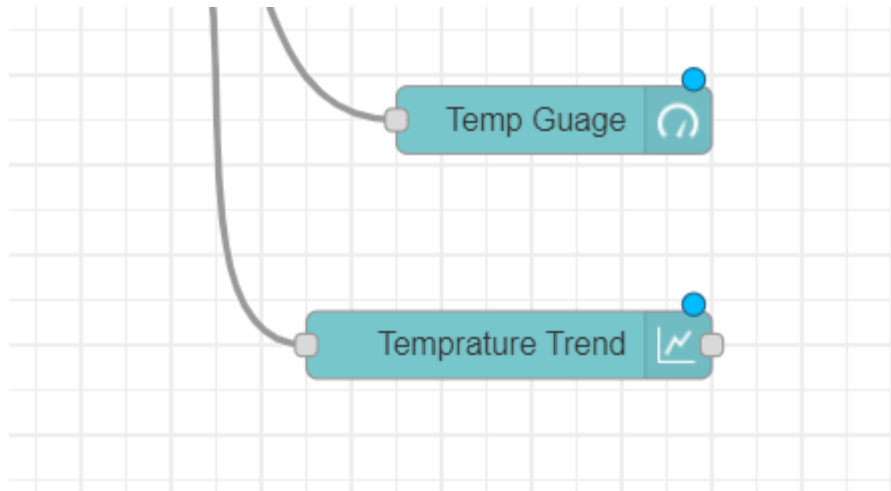
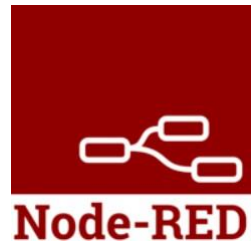
Encrypting the data using RC4 and sending it to cloud using MQTT.
(Python Code)



SECURE Transmission

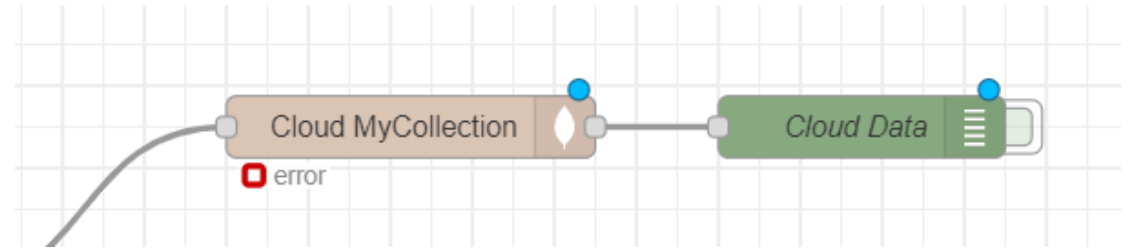


Part 3: Dashboard & Monitoring



node-red-dashboard 3.6.2

Part 4: Cloud Database



node-red-contrib-mongodb-aleph 0.3.0

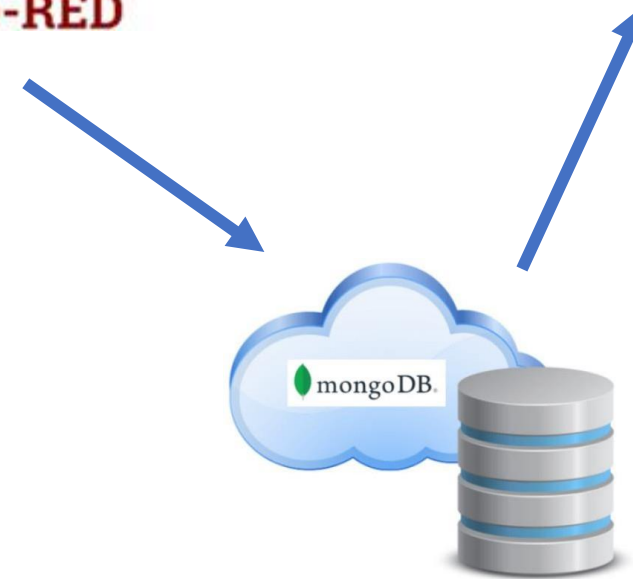
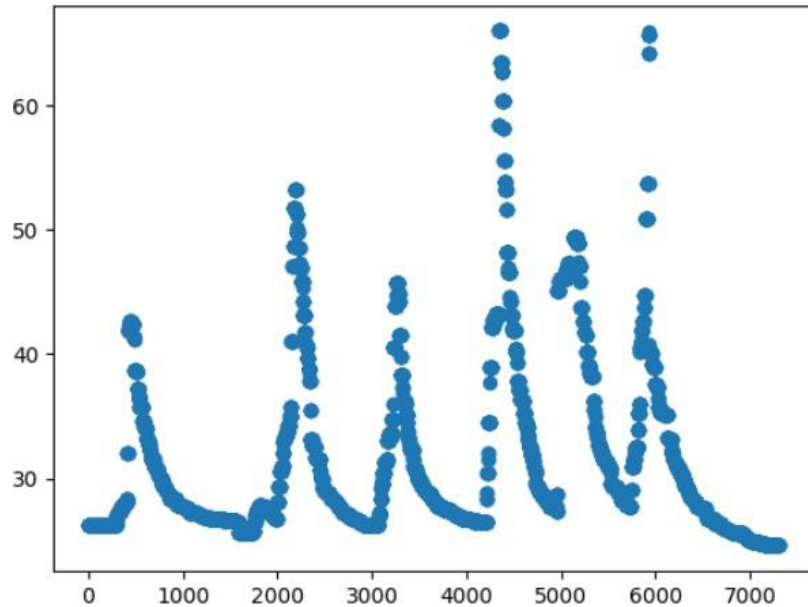
[Filter](#) [Reset](#) [Apply](#) [Options](#)

QUERY RESULTS: 1-20 OF MANY

`_id: ObjectId('65b3d3ad01da125fe019f048')`
`payload: 26.3`
`_msgid: "c8ea51d0d5658175"`

`_id: ObjectId('65b3d3ae01da125fe019f049')`
`payload: 26.3`

Part 4: Cloud Database

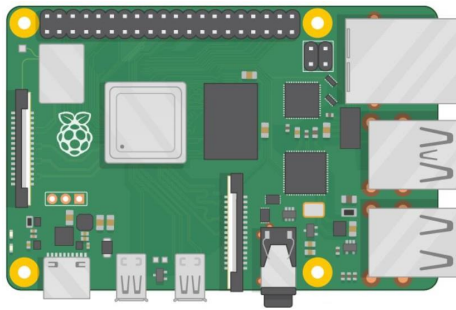
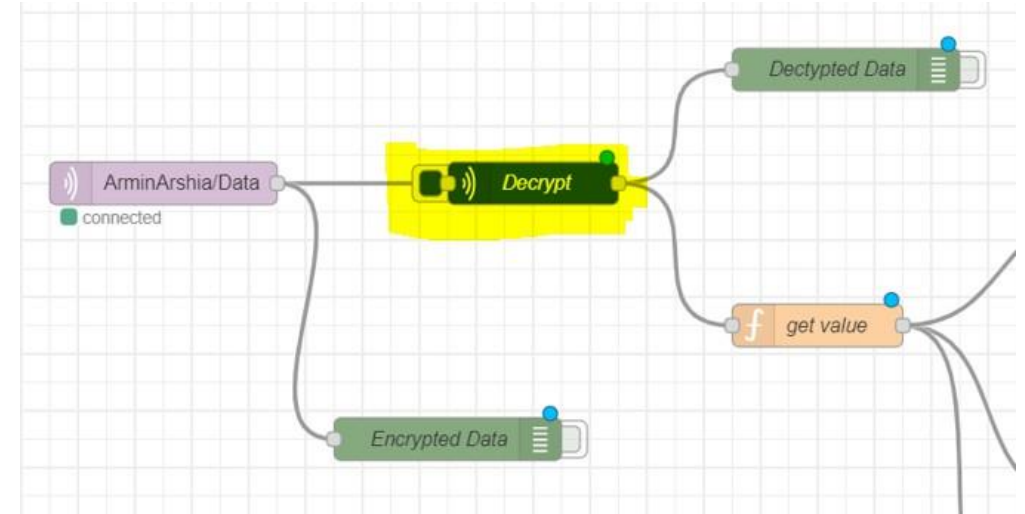


```
# Retrieve temperature data from MongoDB
```

Part 5: Encryption & Decryption

*We use RC4 algorithm for Encryption.
Its implementation is easy but useful for security*

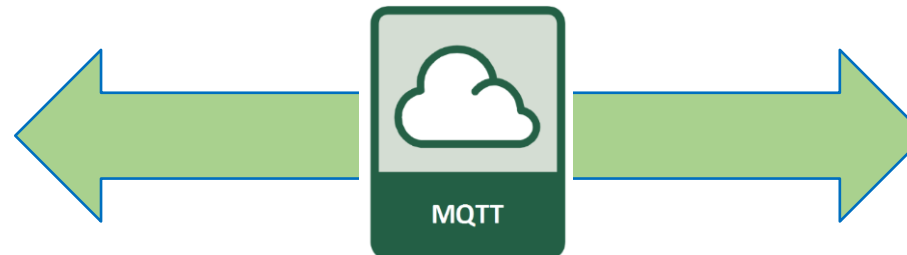
*(Send) Encryption.py --> running on RPI
Data has sent to mqtt broker in secure way
(Receive) Decryption.py --> running Node-red*



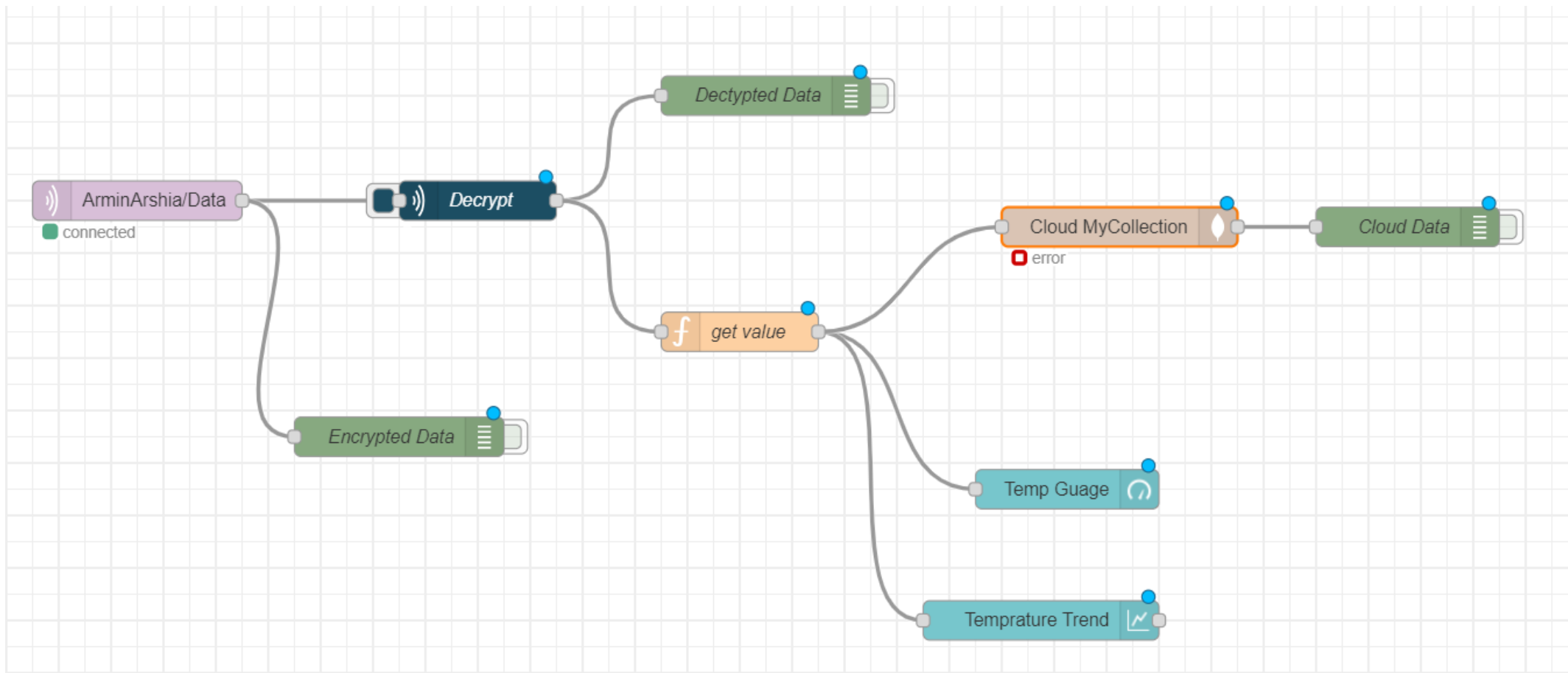
/in: {'temp':35}
/key: Armin

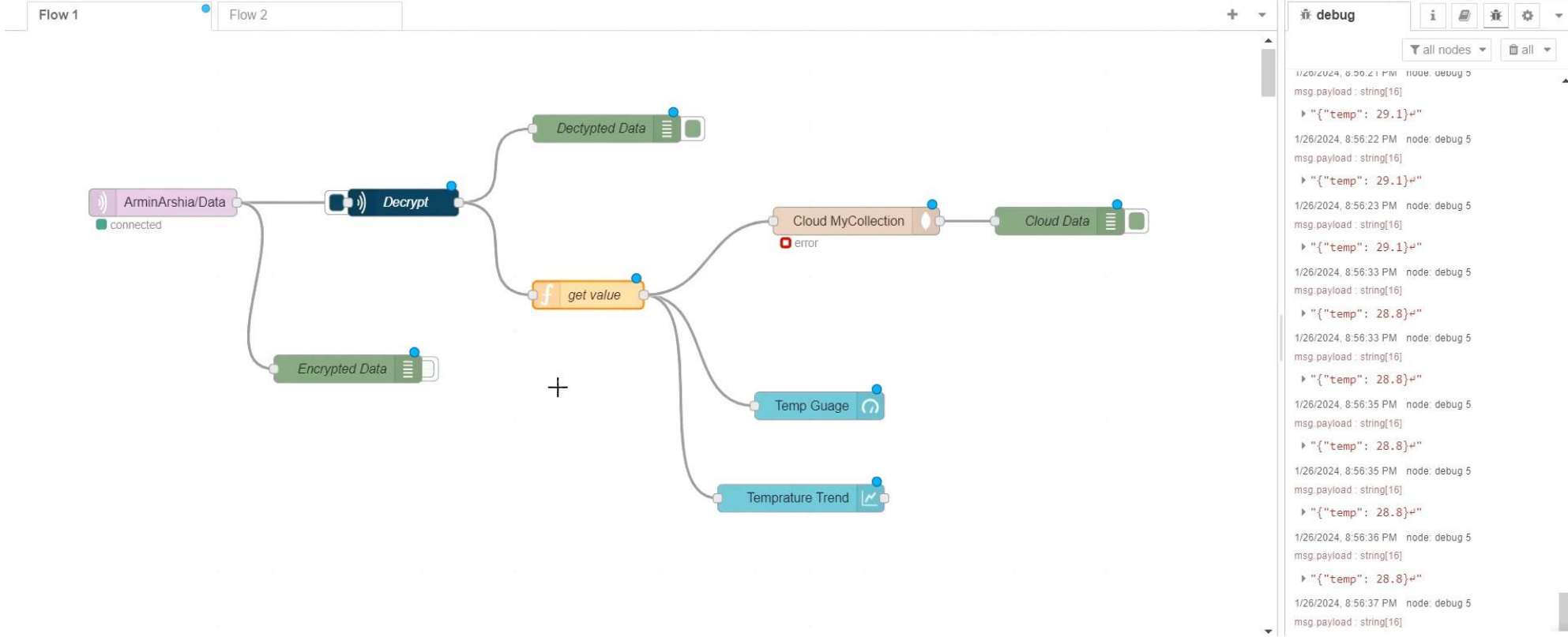
RC4 Encryption

/sent: 4daf9b6d4ba702afcc2497



/out {'temp':35}
/key: Armin





Part 6: (Gateway)

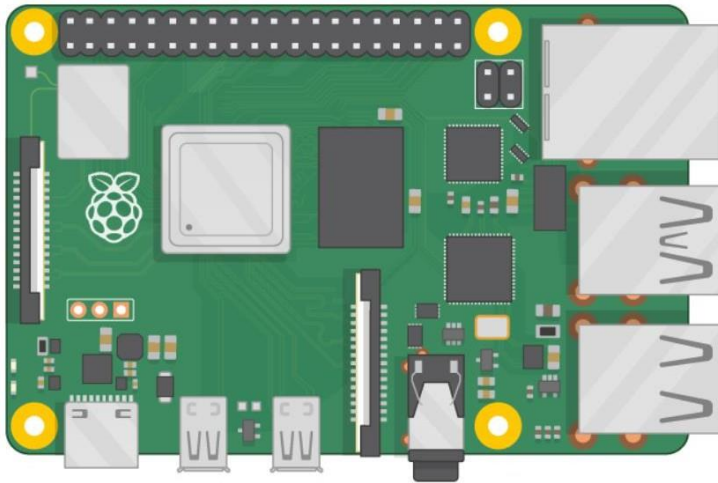
Raspberry pi 4 is used as a gateway.

Tasks:

- Collecting the data from ESP32 via Bluetooth

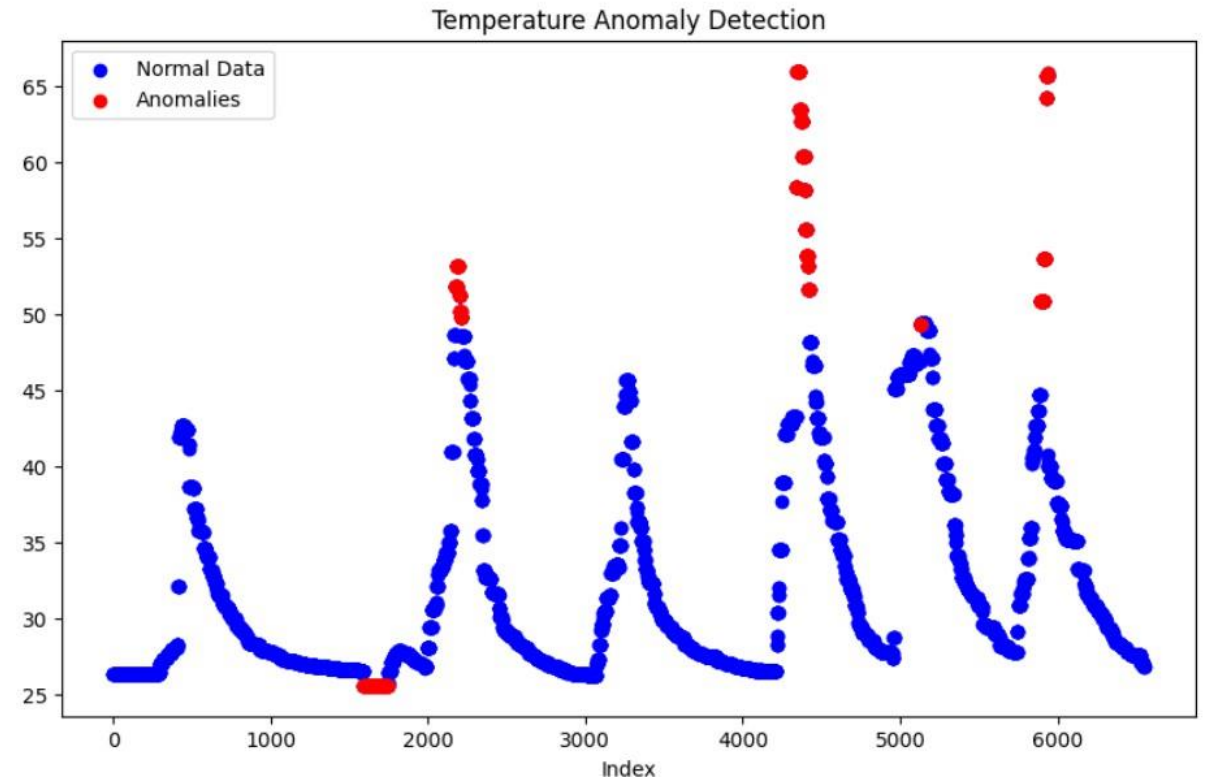
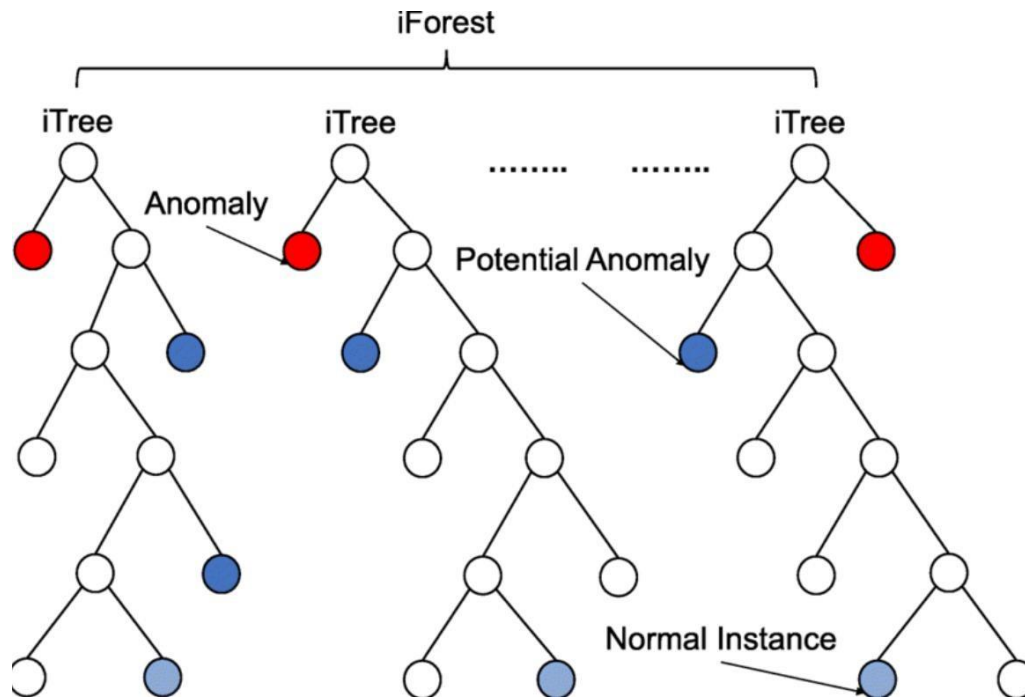
- Encrypting the data using RC4

- Sending the encrypted data to cloud Via MQTT



Part 7: ML Anomaly Detection

Unsupervised ML Algorithm



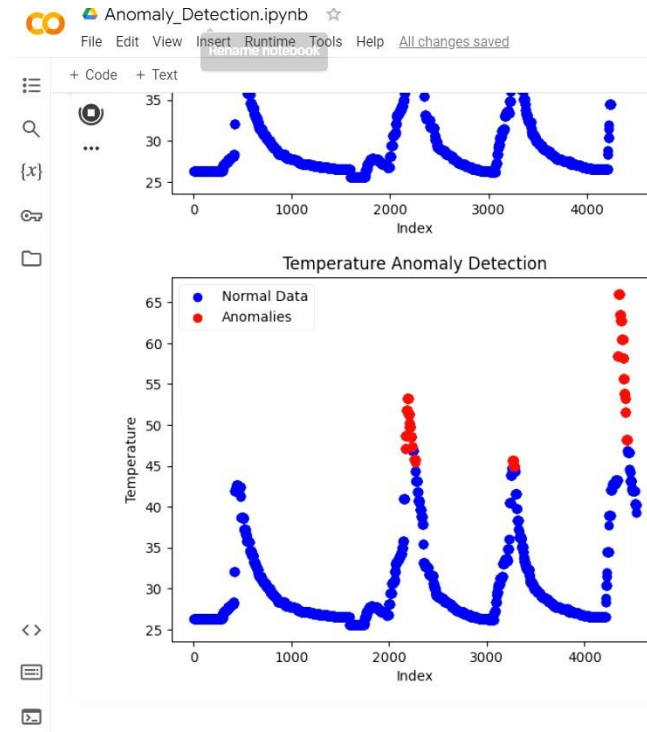
Part 7: ML Anomaly Detection

Unsupervised ML Algorithm

```
while True:
    # Read temperature data
    temperature_data = read_temperature_data()

    # Perform anomaly detection and visualization
    perform_anomaly_detection(temperature_data)

    # Wait for 10 seconds
    time.sleep(10)
```



Thank you !